THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
032 CHEMISTRY 1
MARKING SCHEME APRIL 2025

SECTION A (16 marks)

1. Multiple choice (10marks @ 01mark)

QNS	i	ii	iii	iv	v	vi	vii	viii	ix	x
ANS	D	C	B	B	D	C	E	E	B	A

2. Match the items (06marks @ 01mark)

LIST A	i	ii	iii	iv	v	vi
LIST B	F	B	A	H	I	G

SECTION B (54 marks)

3. (a) Chlorine is a very dangerous gas. It is very poisonous if inhaled to even a small extent. Chlorine poisoning occurs when the gas is inhaled or swallowed. It reacts with water inside and outside of the body (such as water in the digestive tract and moisture on the lungs and eyes) to form hydrochloric acid and hydrochlorous acid. Both of these substances are extremely poisonous.
03 marks

(b) Uses of chlorine

i. Chlorine is bleaching agent and is also used in the manufacture of other bleaches. When chlorine is added to sodium hydroxide solution, bleach is made.

$$Cl_{2(g)} + 2NaOH_{(aq)} \rightarrow NaCl_{(aq)} + NaOCl_{(aq)} + H_2O(l)$$

ii. Chlorine is added to water supplied to homes and industries to kill disease-causing germs like bacteria. It is also used to sterilize the water in swimming pools.

iii. Chlorine is used to make some important chemicals such as hydrochloric acid, chlorofluorocarbons (CFCs), tetrachloromethane (CCl_4), and chloroform ($CHCl_3$).

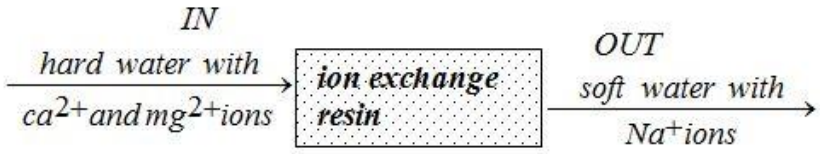
iv. Chlorine is a reactant in the manufacture of organo-chloro compounds which are used to make pesticides, antiseptics (e.g. dettol), herbicides (weed killers) and polyvinyl chloride (PVC) which is an intermediate compound in the production of plastics.

Any four @01½ marks = 06 marks

4. (a) Permanent hardness of water is one which cannot be removed by boiling. Permanent hardness of water is caused by the presence of dissolved calcium sulphate or calcium chloride and magnesium sulphate or magnesium chloride. 03 marks

The common cations (positive ions) found in hard water (Ca^{2+} and Mg^{2+}) enter a water supply by leaching from minerals within an aquifer. Common calcium-containing minerals are calcite and gypsum. A common magnesium mineral is dolomite (which also contains calcium).
03 marks

(b) Ion Exchange:



An ion exchange column removes ca^{2+} and mg^{2+} ions from the water and replaces them with Na^{+} ions

This method removes both temporary and permanent hardness in water by the use of *ion exchange resin.* A typical *ion exchanger* is a container full of small beads. These beads are made of special plastic called ion exchange resin. The resin beads are porous and contain sodium ions. When hard water flows through the resin, the calcium and magnesium ions in the water are exchanged for the sodium ions and attach themselves to the resin. This process, therefore, removes calcium and magnesium ions from the water. They are replaced by sodium ions, which do not make the water hard. (03 marks)

5. (a) Faraday's First Law of Electrolysis states that the mass of a substance liberated at an electrode during electrolysis is directly proportional to the quantity of electricity passing through the electrolyte.

Faraday's Second Law of Electrolysis states that when the same quantity of electricity is passed through solutions of different electrolytes the relative numbers of moles of the elements deposited are inversely proportional to the charges on the ions of each of the elements respectively.

(b) (i) (03 marks)

Current I = 2A

Time t = 9 minutes = 9 × 60 = 540 s

Mass m of metal M liberated was 0.3552 g

Valency v of meta l M = 2

Faraday's constant 96500 C/mol (01 mark)

Relative atomic mass Mr

Mass m liberated $= \frac{MrIt}{vF}$ (01 mark)

$Mr = \frac{mvF}{It}$ (01 mark)

$Mr = \frac{0.3552 \times 2 \times 95600}{2 \times 540}$ (01½ marks)

Mr = 63.5 g/mol (00½ marks)

(ii) The metal is copper (01 mark)

6. (a)

i. fractionating tower/distillation unit (01 mark)
ii. Fractional distillation (01 mark)
iii. Different boiling points (01 mark)
iv.
a) (A) (00½ marks)
(00½ marks)

b) (G)
c) (B) (00½ marks)

v. Used to make Asphalt, bitumen (01½ mark)

(b) Give the IUPAC systematic names of compounds Q and R

P: $CH_2C(Cl)CH_2CH_2CH_3$ **(2-chloro-1-pentene)** (01½ mark)

Q: $CH_3CHClCH_2CH3$**(2-chlorobutane)** (01½ mark)

7. Laboratory Rules

i. Do not enter the laboratory without permission from your teacher or laboratory technician.
ii. Wear safety goggles all the time while in the laboratory.
iii. Never run unnecessarily in the laboratory.
iv. Eating, drinking, and smoking are strictly prohibited in the laboratory.
v. Don't perform any experiment not authorized by your teacher or lab technician.
vi. Never taste anything.
vii. Never use your hands to transfer chemicals. Use a spatula instead.
viii. Notify your teacher or technician immediately in case of an accident
ix. Know what chemicals you are using, carefully read the label twice before taking anything from the reagent bottle. Do not interchange labels.
x. Never point a test tube or vessel that you are heating at yourself or your colleague.
xi. Make sure all the burners are turned off before leaving the laboratory.
xii. When carrying out the experiment where you expect harmful gases to be produced, use the fume chamber. (1 @ = 09 marks)

8. (a). Voumetric analysis is the process of determining unknown concentration of a known compund by tirating it with another solution of known concentration. 001½ mark

(b) Given 98% H_2SO_4 by mass means 98 g H_2SO_4 is present in 100 g of solution.

From the given density, the volume of the concentrated H_2SO_4 is calculated as:

Volume = mass/ density 001½ mark

Hence, volume of the concentrated H_2SO_4 is 100 / 1.84 cm^3 = 54.35 cm^3 = 0.5435 L 001½ mark

The molar mass of H_2SO_4 is 98 g/mol i,e. 98 g H_2SO_4 equals 1 mole of H_2SO_4.

Hence, the molarity of the given concentrated H_2SO_4 is 1 mol / 0.5435 L = 18.4 M 001½ mark

Now, to determine the required amount of concentrated H_2SO_4, the molarity formula is applied as:

$M_1V_1 = M_2V_2$ 00½ mark

Where, M_1 = Molarity of concentrated H_2SO_4 solution

V_1 = Required volume of concentrated H_2SO_4 solution

M_2 = 0.5 M

V_2 = 5 L

Hence, 18.4 M x V_1 = 5 L x 0.5 M 001½ mark

V_1 = 0.136 L = 136 cm^3 00½ mark

Thus, the volume of concentrated sulphuric acid required to make 5 litre of 0.5 M H_2SO_4 solution is 136 cm^3. 00½ mark

SECTION C (30 marks)

9.

Introduction: Metal extraction is the process of separating and isolating metals from their ores, which are naturally occurring minerals containing metal compounds. 01½ marks

There are four main stages for the extraction of moderately reactive metals:

1. Mining and concentration of the ore:

   The unwanted rock material in the ore is called **gangue**. The ore is treated to remove as much gangue as possible. For example:

   - It is crushed and washed., andthen grinded down to fine powder.
   - A method called **froth flotation** is used with sulphide ores (e.g. CuS or ZnS). "frothing" chemicals (a suitable oil) are added, Sulphides are attracted to these chemicals. When air is blown through the slurry, froth rises to the top of the tank carrying the metal sulphides with it.
   - Magnetic separation can be used.

2. Roasting in air 03 marks

   For the ores that occur as sulphides or carbonates of the metal, the concentrated ore is heated (roasted) in air to convert the ore into an oxide, for example:

   1.$2PbS_{(s)} + 3O_{2(g)} \rightarrow 2PbO_{(s)} + 2SO_{2(g)}$

   2. $2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)}$

   3. $ZnCO_{3(s)} \rightarrow ZnO_{(s)} + CO_{2(g)}$

   Most copper is extracted from copper pyrites, $CuFeS_2$. The ore is roasted in a limited supply of air to produce to produce copper (I) sulphide:
   $2CuFeS_{2(s)} + 4O_{2(g)} \rightarrow Cu_2S_{(s)} + 3SO_{2(g)} + 2FeO_{(s)}$
   The copper (I) sulphide is then reduced to copper by heating in air: $Cu_2S_{(s)} + O_{2(g)} \rightarrow 2Cu_{(s)} + SO_{2(g)}$
   The copper produced by this method is never pure. It must be refined (purified) by electrolysis if it has to be used for electrical wiring. 03 marks

3. Reduction of oxides to metals

   This is another important stage in the extraction of metal. Here, the roasted ore must now be reduced to respective metals. Reduction occurs at a very high temperature. The materials employed for reduction are mainly carbon or carbon monoxide. Thus; 03 marks
   $Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(l)} + 3CO_{2(g)}$
   $ZnO_{(s)} + C_{(s)} \rightarrow Zn_{(s)} + CO_{(g)}$

4. Purifying the metal

   Purification is normally done through a number of ways, which include: 03 marks

   - *Electrolysis*: Electrolysis is used to produce a pure metal directly from its molten compounds…
   - *Oxidation*: The molten crude metal is exposed to hot air in a furnace. The impurities in the crude …
   - *Distillation*: In distillation, the crude metal is heated in a furnace until the pure metal evaporates, leaving behind the impurities. The vapour is then collected and condensed in a separate chamber.
   -

Conclusion: 01½ marks

Total: 15 marks

10. Importance of the ability of matter to change states.
Matter is... and exists in three states… (Intro: 01½ marks)

The Importance of Changing One State of Matter to Another

1. Separation of mixtures

It enables separation of mixtures can through such processes as distillation, sublimation, evaporation and condensation.

2. Industrial manufacture of products

Industrially, the process of distillation is applied in the production of pure substances such as beer and other alcoholic drinks such as wine. The manufacturing process involves boiling, evaporating and condensation.

3. Refining of petroleum (crude oil)

Crude oil contains organic liquid components, each with a different boiling point. In the refinery, the components with lower boiling points evaporate first and get separated out, leaving those with higher boiling points behind. In this way, we get various types of oil components (fractions) such as petrol, diesel, kerosene, lubricating oil, etc.

4. Drying of crops and clothes

When you suspend your clothing on a cloth line to dry, the moisture in it is lost through evaporation. Likewise, farmers in the village often spread crops on the ground to dry.

5. Cooling of our bodies in hot weather

Cooling of our bodies in hot weather depends in the evaporation of sweat or water.

6. Ice formation in refrigerators

You can freeze water into ice and then use the resulting ice for cooling the beverage or preserving foods.

7. Melting metals to make alloys

In metallurgical industries, need may arise to mix two or more metals (alloys) together. This is only possible, where two or more metals are first melted at high temperatures into liquids. Then the resulting liquid metals are mixed in appropriate proportions. This is followed by cooling down the mixture to a solid alloy.

(Any six @02 = 12 marks)

8. Formation of rain

Rain is mainly formed through the process of evaporation and condensation. Water vapour, evaporating mostly from water bodies (oceans, seas, lakes, rivers, ponds, etc), land and plants rises up to the sky. As it rises, it cools down and condenses into tiny droplets

Conclusion:

Total: 15 marks

11. Global warming

Introduction: Global warming is the increase in the average temperature of the earth's atmosphere as a result of the greenhouse effect. (01½ marks)

Global warming mainly results from human activities such as fossil fuel burning and deforestation, which produce greenhouse gases such as carbon dioxide and some oxides of nitrogen (NOx). Carbon dioxide is responsible for over 50% of the global warming. (03 marks)

The greenhouse gases bring about greenhouse effect in the atmosphere, which allow ultraviolet radiations to pass through them and reach the earth. As the earth's surface gets heated up, some of the heat is radiated back to the atmosphere. A layer of the greenhouse gases in the atmosphere, acting as a blanket, prevents the heat from escaping to the upper atmosphere. This causes excess heat in the air around the earth (atmosphere), hence the phenomenon called global warming. (04½ marks)

Effects:

- Global warming may result in melting of the polar ice caps, causing an increase in levels of the ocean water. This may lead to flooding of the coastlands and islands.
- Other effects include increase in the average temperature of the earth, (04½ marks)
- change in weather patterns, and increased desertification, which would in turn cause reduction of the arable land.

Conclusion: Any relevant (01½ marks)

(Total 15 marks)